UBER TRIP ANALYTICS

Analyzing Urban Mobility, Demand Patterns and Operational Performance

1. Problem Statement

Ride-hailing services such as Uber generate large volumes of trip data across different times, locations, vehicle types and payment methods. Understanding this data is important for improving service availability, allocating vehicles efficiently and identifying patterns in customer demand and revenue.

This project analyzes Uber trip records to identify when and where demand is highest, which routes and locations are most frequently used, how performance differs across vehicle types, and how fares and surge fees vary. Python is used for data cleaning and preparation, SQL for answering business questions, and Power BI for interactive visualization.

2. Project Objectives

- Identify daily and hourly trip-demand patterns.
- Find the most popular pickup and drop-off locations and frequently travelled routes.
- Compare trip volume, revenue, average fare and trip duration across vehicle types.
- Understand payment-method preferences and their contribution to revenue.
- Examine surge-fee patterns and the relationship between trip distance and fare.
- Present findings through an interactive Power BI dashboard.

3. Project Scope

Data preparation: Clean trip and location data, handle invalid records, standardize fields, calculate trip duration and join location details.

Data analysis: Use SQL queries to investigate demand, revenue, locations, routes, vehicle performance, payment methods and surge pricing.

Visualization: Build a Power BI dashboard with KPI cards, charts and slicers for exploring trip date, payment type, pickup city and vehicle type.

Tools: Python (Pandas), MySQL, Power BI and Google Colab.

4. Expected Outcomes

- A cleaned and joined dataset suitable for analysis.
- SQL-based findings that answer the project's business questions.
- An interactive dashboard showing key trip, demand and revenue patterns.
- Actionable recommendations for improving vehicle allocation, service availability and operational planning.

Project focus: Descriptive and diagnostic data analytics. Machine-learning prediction is outside the scope of this project.